

MIRAVA POLYSCOPE

Imaging across all scales – from millimeters to 3.2 nanometers



Sample preparation information

Prepare your own samples for **STED** super-resolution imaging with your *abberior MIRAVA* microscope.

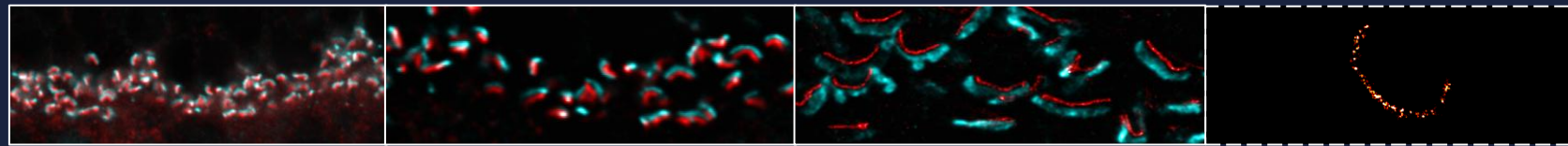
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MIRAVA POLYSCOPE – one for all

Confocal, *MATRIX*, STED, and *MINFLUX*



mm – 200nm

200 – 60nm

60 - 25nm

25 – 3.2nm

- Confocal

optionally:

- *RAYSHAPE*
adaptive optics
- *TRUESHARP*
deconvolution
- *MATRIX*
array detection

- *2x MATRIX* array

optionally:

- *TRUESHARP*
deconvolution
- *gentle STED*
(+lifetime)

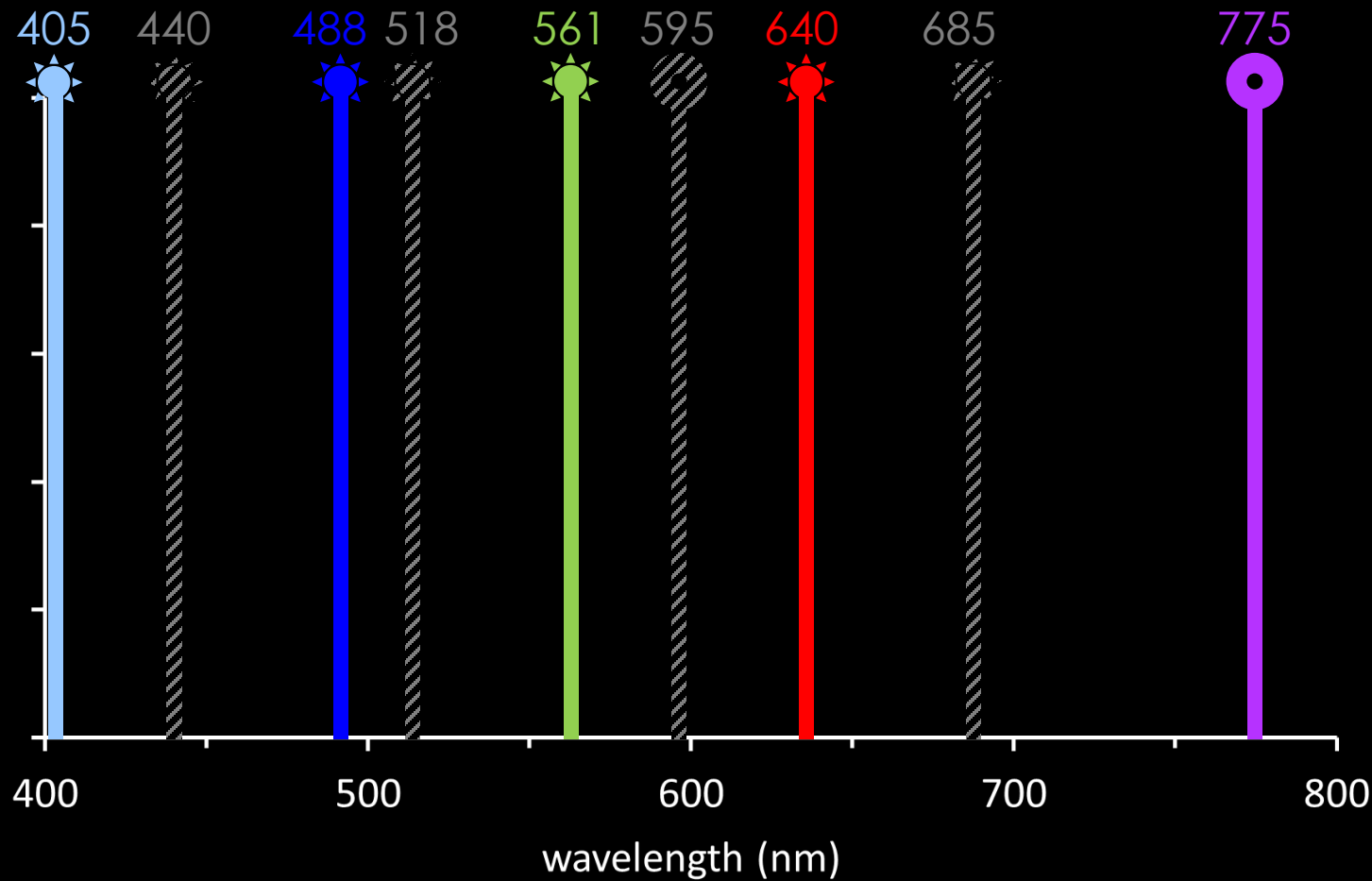
- STED

optionally:

- *TIMEBOW*
lifetime imaging

- *MINFLUX*
(not installed)

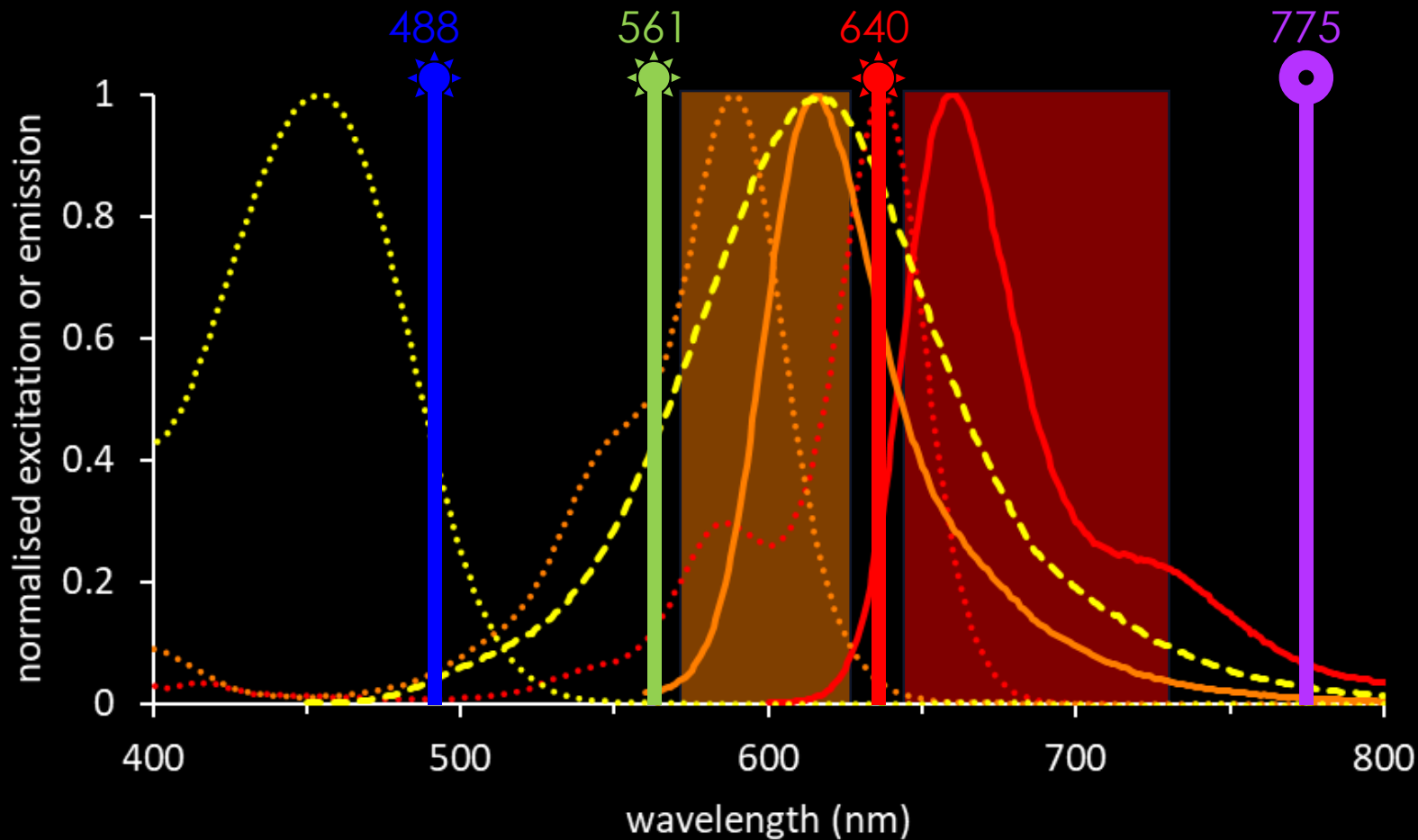
Multi-colour: Lasers



Available lasers

- Excitation
 - 4x installed
 - Upgradable to 7x
- STED lasers
 - 1x installed
 - Upgradable to 2x
- MINFLUX
 - Not installed
 - Upgradable 640 nm

3-colour: Long Stokes STED 775 nm



3-colour (LS)

- STED 755 nm pulsed
- Red dyes
- Orange dyes
- Long Stokes shift dyes
- Use ca. 2x higher STED laser power for same resolution
- + High resolution
- + Low-med bleaching
- + Single step
- + Perfectly aligned image channels
- + Good spectral separation
- Works better with 440 nm excitation (upgrade)

e.g. **STAR RED** & **STAR ORANGE** & **abberior STAR 460L**

3-colour: Long Stokes STED 775 nm

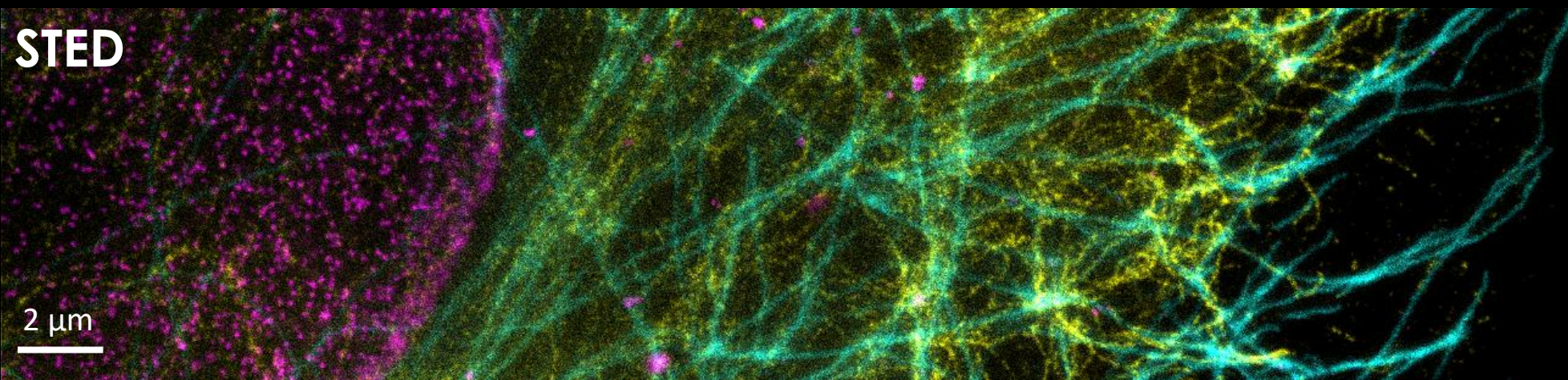
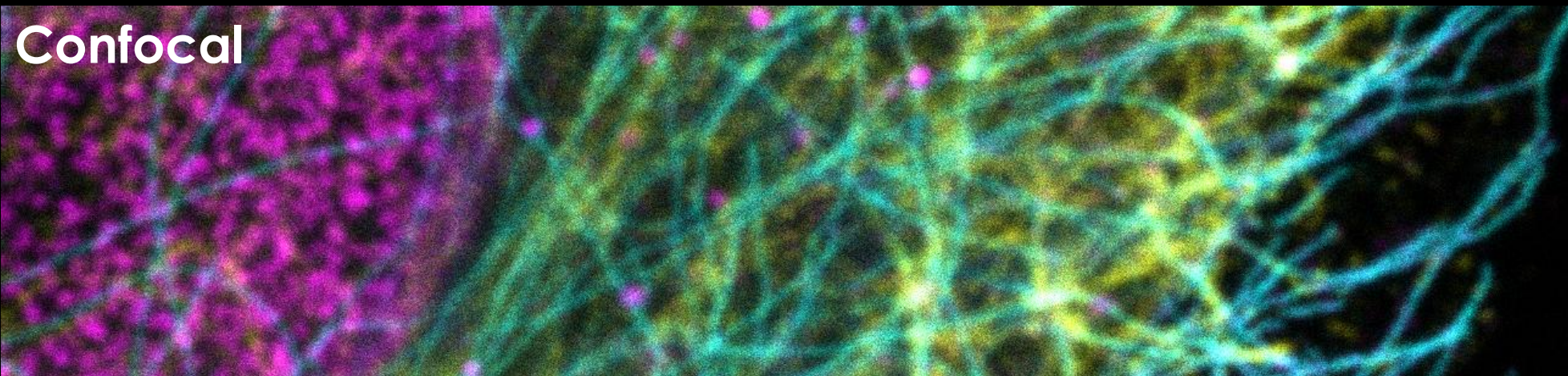


Improve channel separation with your current microscope configuration:

- **STAR RED** & **STAR ORANGE** & **abberior STAR 460L**
 - Adjust detection: 650-730 nm instead of 571-630 nm.
 - Less crosstalk with STAR ORANGE (but lower signal)
 - Implement by selecting ATTO 490LS in software
- **STAR RED** & **STAR ORANGE** & **ATTO 490LS**
 - ATTO 490LS more red-shifted than STAR 460LS.
 - ATTO 490LS fits your 488 nm laser better.
 - Automatic detection around 650-730nm
= reduced crosstalk with STAR ORANGE.
 - Limited range of ATTO 490LS commercial conjugates

Tubulin STAR 460L
Nuclear pore STAR RED
Vimentin STAR ORANGE

***3-colour:
Long Stokes
STED 775 nm***



STED: Popular dyes for fixed samples

STAR, LIVE & FLUX dyes made by abberior + other commercially produced dyes
Many are available as large range of conjugates: Antibodies, toxins...

Deep red dyes (emission 690-780 nm)

STAR deepRED, CF680R

CF680*, FLUX 680* (untested: ATTO 680, Alexa 680)

Red dyes (emission 650-690 nm): Best resolution

STAR RED & 635P & 635, ATTO 647N,

ATTO 633 & 643 & 655, CF633 & 640 & 640R*, JFX650, JF646 & 669, SiR, Alexa 633 & 647*, Quasar 670*, Cy5*

Orange dyes (emission 570-630 nm): 2nd best resolution

STAR ORANGE, STAR 580, Alexa 594

ATTO 594, ATTO 590, CF594, JF549, Alexa 546*, Alexa 568*, Quasar 570*, Cy3*

Long Stokes shift dyes (emission 570-690 nm)

STAR 460L,

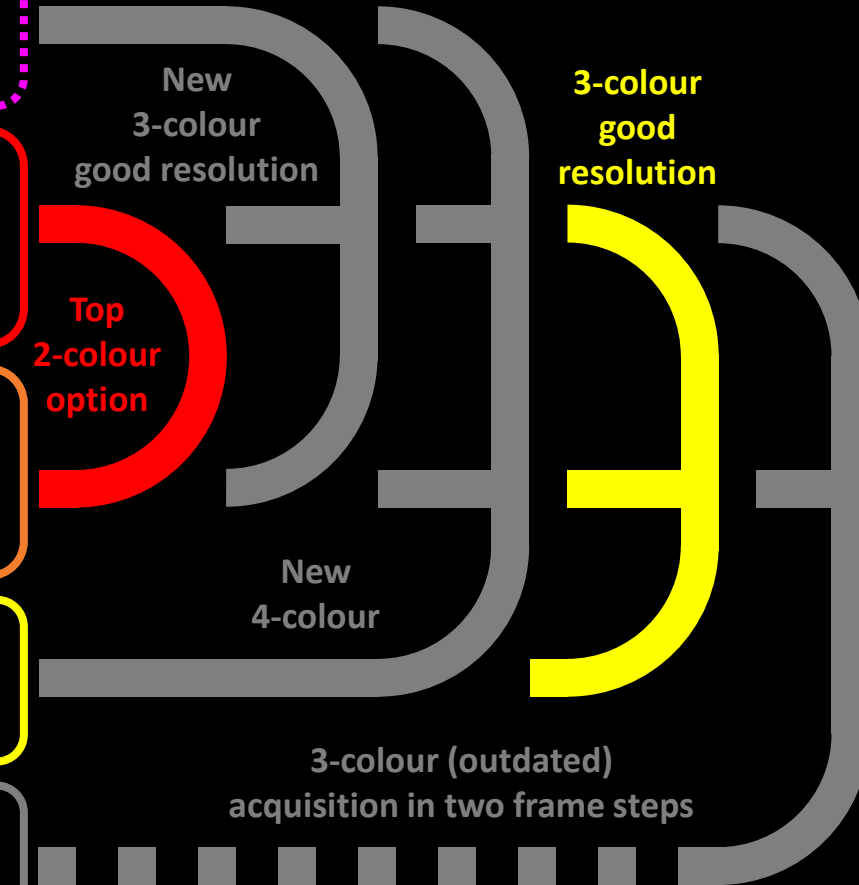
ATTO 490LS, Abberior STAR 520L*

Green dyes (emission 500-550 nm)

STAR GREEN, Oregon Green 488, Alexa 488, ATTO 488, STAR 488, FITC, Chromeo 488 & 505, DyLight 488

Long Stokes shift dyes (emission 500-550 nm)

STAR 440L*



* Dyes may show increased photo-bleaching with STED

--- If excited at 640 nm, deep red show crosstalk with red dyes

Greyed-out: Optional laser upgrade (not installed)

STED 775

STED 595

STED: Dyes for live-cell

STAR, LIVE & FLUX dyes made by abberior + other commercially produced dyes

Red dyes (emission 650-700 nm): Best resolution

Silicon Rhodamine (SiR), LIVE 610, LIVE RED, JFX650, JF669 & 646, SPY650 & 620, FM4-64* propidium iodide*

Membrane stains:

STAR RED membrane, DiD, MemBrite Fix640-660

Mitochondria stains:

LIVE RED mito, MitoBright LT Deep Red*, Nile Blue*/***

Orange dyes (emission 570-630 nm): 2nd best resolution

LIVE 550 & 560 & 590, JF549, ATTO 590, ATTO 594, SPY 595, Alexa 594****

Membrane: STAR ORANGE membrane

*Mito: LIVE ORANGE mito⁺, PK Mito Orange⁺, Nile Red****

Long Stokes shift dyes (emission 570-690 nm)

LIVE 460L

Green dyes (emission 500-550 nm)

LIVE 510

SPY 505

Pico Green (DNA)

FITC

3-colour
good
resolution

Top
2-colour
option

3-colour (outdated)
two-step acquisition

Many live-cell dyes are available as

- **direct stains**
for tubulin, actin, DNA, lysosomes, mitochondria...
- **substrates for protein tags:**
SNAP, CLIP, HALO, HALOX
- **other conjugates**
with proteins or metabolites (insulin, lectins...) + custom conjugates via + dye-NHS esters

* Dyes may show increased photo-bleaching with STED

** Surface stain only (cell-impermeable)

*** Also stains lipid droplets, slightly ER, plasma membrane

+ top mitochondria dyes

Greyed-out: Optional laser upgrade (not installed)

STED 775

STED 595



Samples for STED



Do I need special samples for STED ?

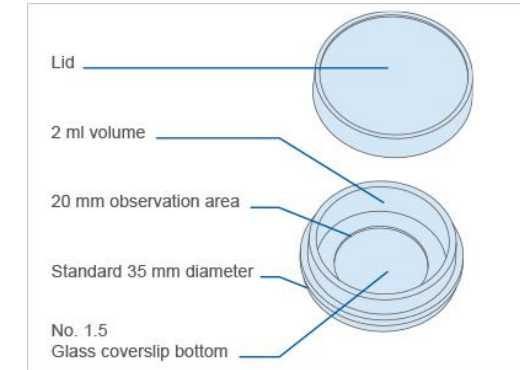
General requirements



Requirements are similar to confocal microscopy.

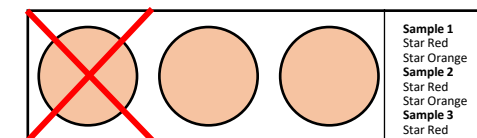
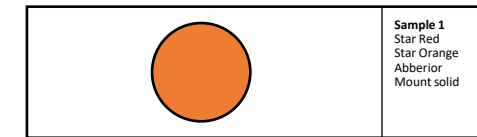
Dishes and slides

- Use glass bottom petri dishes, slides, chamber slides...
- Thickness #1.5 = 0.17 mm
- No plastics substrates
- No gridded cover slips
- Slides: Avoid mounting on slide edges to fit sample holder



Embedding (optional)

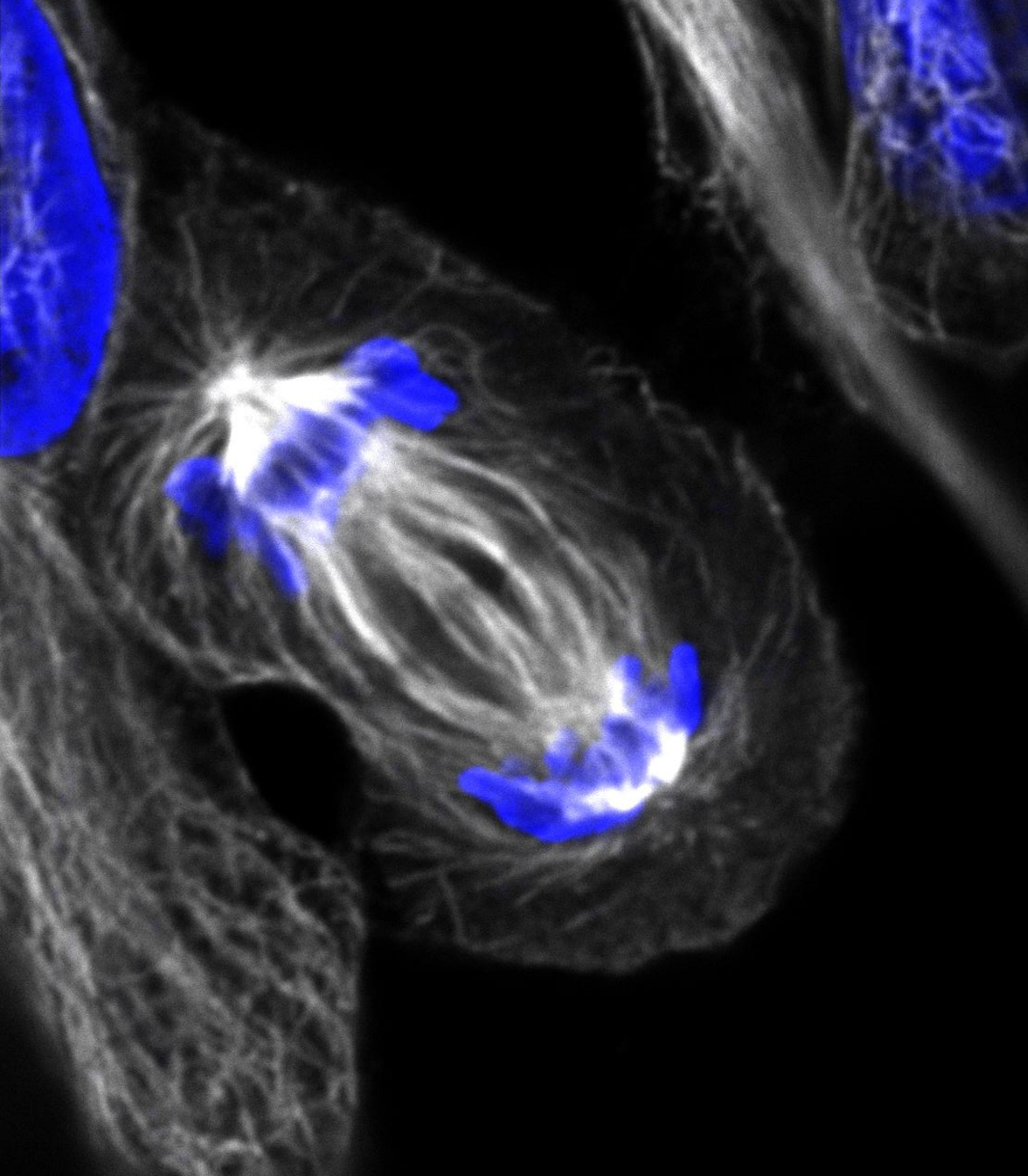
- Mounting media with antifade help getting more light out of fixed samples.
- The better the RI of the mounting medium matches the immersion liquid of the objective – the deeper the imaging



Protocols and further infos

- <https://abberior.rocks/expertise/protocols/>





STED: What about... DAPI ?



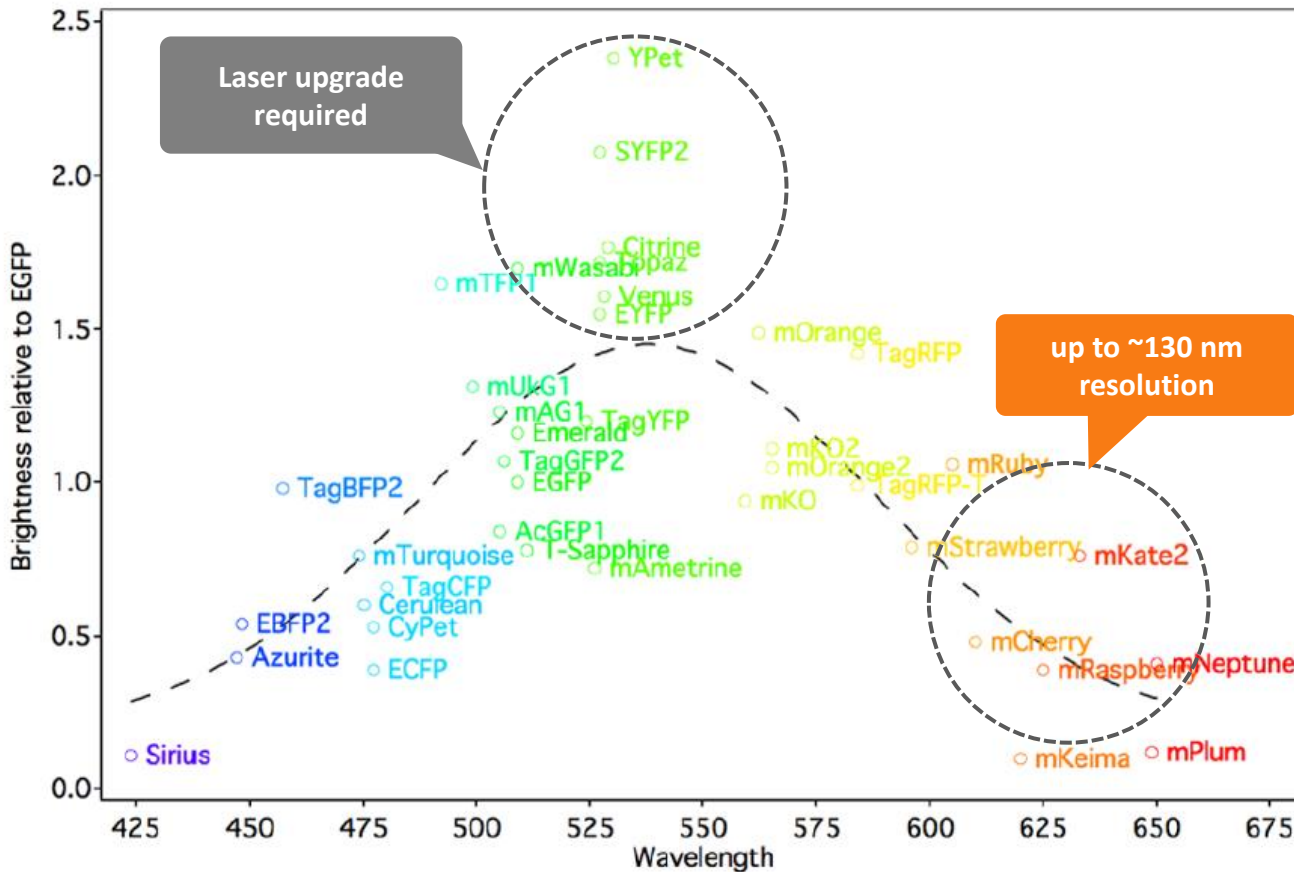
DAPI (excitation 405 nm)

- Only in **low concentrations** (~0.5ug/ml)
- Use a **wash step** after DAPI stain
- Please avoid DAPI in mounting media
- Same applies to Hoechst staining

High DAPI concentration can result in additional background through 2-photon excitation by the STED laser



STED: What about... fluorescent proteins?



FPs generally show lower STED performance than organic dyes

→ Use FPs only if organic stains are unavailable

→ In fixed samples, immuno-stain FPs for increased resolution

- **Green FPs** are typically superior to **red FPs** but require the STED 595 nm upgrade.
- **Red FPs** like mCherry or tagRFP:
 - Up to ~130 nm STED resolution.
- **New red FP variants** may offer better performance:
 - Try mNeptune2
 - Try mScarlet3

Objective lenses VS mounting media



Objective		Refractive index (RI)	Sample type for deep imaging
Oil		>1.52	Fixed samples in: Methyl Salicylate, Ethyl Cinnamate
		~1.52	Fixed samples in: ProLong Glass, TDE-O, (RapiClear1.52)
		<1.5	Fixed samples in: Abberior Mount Solid AF, Mowiol+Dabco, ProLong Gold/Diamond, (RapiClear1.49/1.47)
Glycerol		~1.45	(Objective not available) Fixed samples in: Glycerol
Silicone		~1.4	(Requires objective upgrade) Dense tissue in aqueous buffer Fixed samples in TDE-S, Glycerol mix, Abberior Mount Liquid AF
		<1.4	(Requires objective upgrade) Less dense tissue in aqueous buffer Fixed samples in Glycerol mix (low %)
Water		~1.3	(Requires objective upgrade) Cultured cells or less dense tissue samples (live or fixed) in aqueous buffer, live plant tissue
Air		<1.2	(Requires objective upgrade) Mostly non-biological samples



Mounting media for STED & confocal



Hardening media

Non-hardening (liquid) media

Aqueous buffer



All-rounder preferred for most samples

- + High stability & long storage
- Shrinkage: Sample becomes flatter
- + Easy mounting procedure & no sealing required
- Best with oil immersion objectives
- Use versions with antifade for brighter images and longer storage.
- Examples: abberior Mount Solid AF, ProLong Gold, Mowiol + DABCO
- Advanced examples for deeper imaging: ProLong Glass



Preferred for plates & short-term storage

- No stabilisation
- + Sample shape maintained
- Sealing required to avoid slippage or evaporation (use nail varnish, Twinsil or epoxy for coverslip sealing)
- For oil or silicone objectives
- Use versions with antifade for brighter images and better storage.
- Examples: abberior Mount Liquid AF, buffered glycerol + DABCO,
- Advanced examples for deeper imaging: TDE*, Methyl salicylate**, RapiClear***



Mostly for live-cell or special applications

- No stabilisation
- + Sample shape maintained
- Sealed dishes or sealed coverslips required (for live-cell best seal coverslips with Twinsil)
- Best with water immersion objective
- Usually no antifade: Faster bleaching, only short-term storage
- Examples: PBS, culture medium (without indicator dye)
- Advanced examples: Distilled water (for ExM+STED microscopy)

Deeper imaging with oil objectives: Suitable mounting media



Hardening media with antifades

- Abberior Mount Solid AF, ProLong Gold, Mowiol+DABCO
- RI match ++: 3D-STED 20-40 μm depth
- Shrinkage during curing: Structures appear flatter

ProLong Glass (solid) / SlowFade Glass (liquid)

- RI match +++: Deeper 3D-STED
- Good antifade but more STED power needed
- Solid with low sample shrinkage / liquid without shrinkage
- Spectral changes in some orange dyes

Abberior Mount Liquid AF

- RI match +: 3D-STED 10-20 μm depth
- No sample shrinkage

RapiClear 1.52

- RI match +++++: Very deep 3D-STED
- Fast tissue clearing effect
- Increased STED bleaching for many dyes

TDE

- RI match +++++: Very deep 3D-STED
- Tissue clearing effect for deeper imaging
- Complex protocol
- Not for fluorescent proteins or Phalloidin

Methyl salicylate

- RI match +++++: Very deep 3D-STED
- Tissue clearing effect for deeper imaging
- Complex protocol
- Tested with STAR RED and STAR ORANGE

SeeDB, Cubic

- Popular in neuroscience
- Good results in kidney tissue
- Cubic: Good clearing and STED performance



Deeper imaging with silicone objectives: Suitable mounting media



Live tissue

- RI match +++: Deep 3D-STED
- Silicone objective is suitable for live brain and dense tissue
- Tissue scattering can reduce signal and resolution

Fixed tissue in aqueous buffer

- RI match ++: 3D-STED to 20-40 μm depth
- Additional tissue clearing important.

Liquid media with antifades

- RI match +++++: Very deep 3D-STED
- Abberior Mount Liquid AF
- Glycerol mix (ideally refractive index near 1.4)
- No shrinkage
- Additional tissue clearing may be required

TDE (premix for silicone objectives)

- RI match +++++: Very deep 3D-STED
- Tissue clearing effect for deeper imaging
- Complex protocol
- Not for fluorescent proteins or Phalloidin

Hardening media with antifades

- Abberior Mount Solid AF, ProLong Gold, Mowiol+DABCO
- RI match ++: 3D-STED 20-40 μm depth
- Shrinkage during curing: Structures appear flatter

**--- Objective not installed ---
Contact us for an upgrade.**

Deeper imaging with water objectives: Suitable mounting media



Live tissue

- RI match ++: 3D-STED depth depends on density
- Water objective is suitable for less-dense tissues
- Tissue scattering can reduce signal and resolution

Fixed tissue in aqueous buffer

- RI match +++++: Very deep 3D-STED
- Additional tissue clearing important.

Liquid media with antifades

- RI match +: 3D-STED 10-20 μm depth
- Abberior Mount Liquid AF
- Glycerol mix (ideally low glycerol content)
- No shrinkage
- Additional tissue clearing may be required

ExM samples in distilled water

- RI match +++++: Max. depth 3D-STED
- Typically extremely clear samples:
Extremely deep 3D-STED
- Complex protocol
- Additional resolution gain:
Combined effect ExM * STED

**--- Objective not installed ---
Contact us for an upgrade.**



Your samples for STED

Cells on coverslips

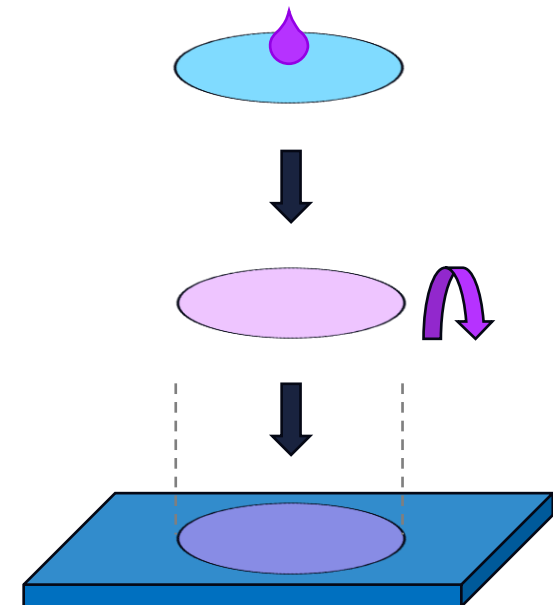


Sample type

- Cultured cells, fixed, permeabilised & stained (see previous pages for dyes)

Mounting

- Solidifying mounting medium for stability during shipping and under intense laser light:
e.g. **abberior MOUNT SOLID AF**
- Add small amount mounting medium onto coverslip
- Then mount coverslip directly on glass slide (cells facing towards slide).
- Alternatively, use well slides or dishes with a glass bottom (thickness #1.5).
- Leave to cure overnight in a dark place (hardening).
- Fix or seal coverslip edges with nail varnish.



Cells on membrane substrate

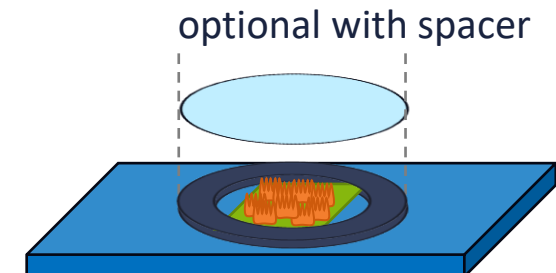
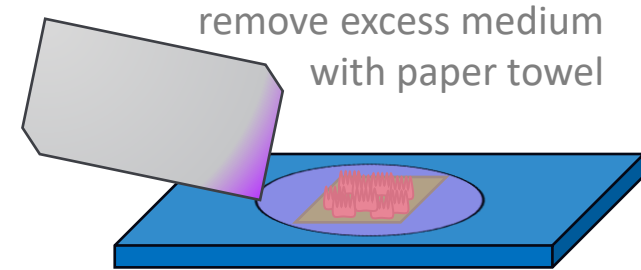
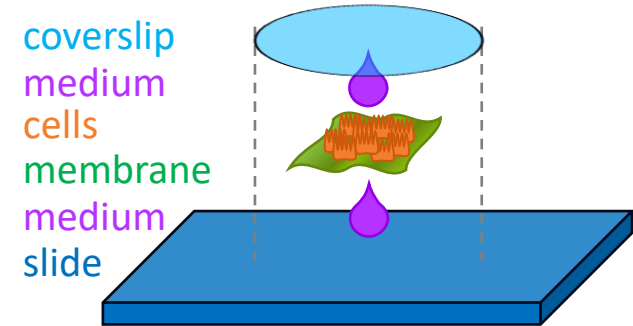


Sample type

- Cells cultured on a membrane substrate, fixed, stained (see previous pages for dyes)

Mounting suggestions

- Mount with glass coverslip (thickness #1.5) and slide.
 - Cells should face towards glass coverslip.
 - Optional: Spacer with appropriate thickness (e.g. from [SUNJin Lab](#))
- Minimise gaps between coverslip and membrane:
 - Sandwich coverslip/membrane/slide using small mounting medium volume (10-20 μ l).
 - Remove excess mounting medium using a paper towel at the edge of the coverslip.
 - Spacers may increase the gap (test required thickness)
 - If gap remains $>50 \mu$ m, optimise refractive index of mounting medium (see previous slides)
- Solidifying mounting medium (best option for shipping):
 - e.g. **MOUNT SOLID AF**
 - Leave to cure overnight in a dark place. Then fix coverslip with nail varnish.
- Liquid mounting medium (less sample shrinkage):
 - e.g. **MOUNT LIQUID AF**
 - Coverslip edge needs to be sealed, e.g. with nail varnish.
- Aqueous mounting buffer: Less stable for shipping, may help for very large gaps ($>75 \mu$ m).



Spheroids / Organoids



Sample type

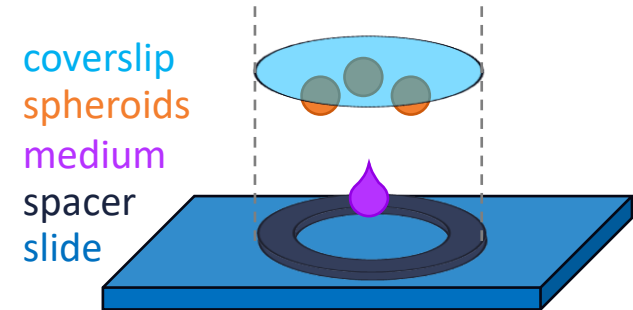
- Spheroids and organoids, fixed, permeabilised, stained

Stains

- Easy STED super-resolution test: abberior STAR RED Phalloidin
- Check previous pages for a large choice of stains and dyes (e.g. immunostaining)

Mounting suggestions

- **Option A:** Mount with glass coverslip (thickness #1.5) and slide
 - Recommended: Spacer with appropriate thickness (e.g. from [SUNJin Lab](#))
 - Solidifying medium (preferred for shipping) or liquid mounting medium:
e.g. **MOUNT SOLID AF** or **MOUNT LIQUID AF**
 - Minimise gap between coverslip/spheroids: Appropriate spacer thickness or multiple spacers
 - If using solidifying mounting medium: Cure (solidify) overnight in a dark place.
 - Fix or seal coverslip edges with nail varnish (or use sticky spacers).
- **Option B:** Mount in multi-well slide or dish
 - Needs to have a glass bottom (thickness #1.5)
 - Solidifying medium prevents movement and spills during shipping:
e.g. **MOUNT SOLID AF**
 - Cure (solidify) overnight in a dark place.

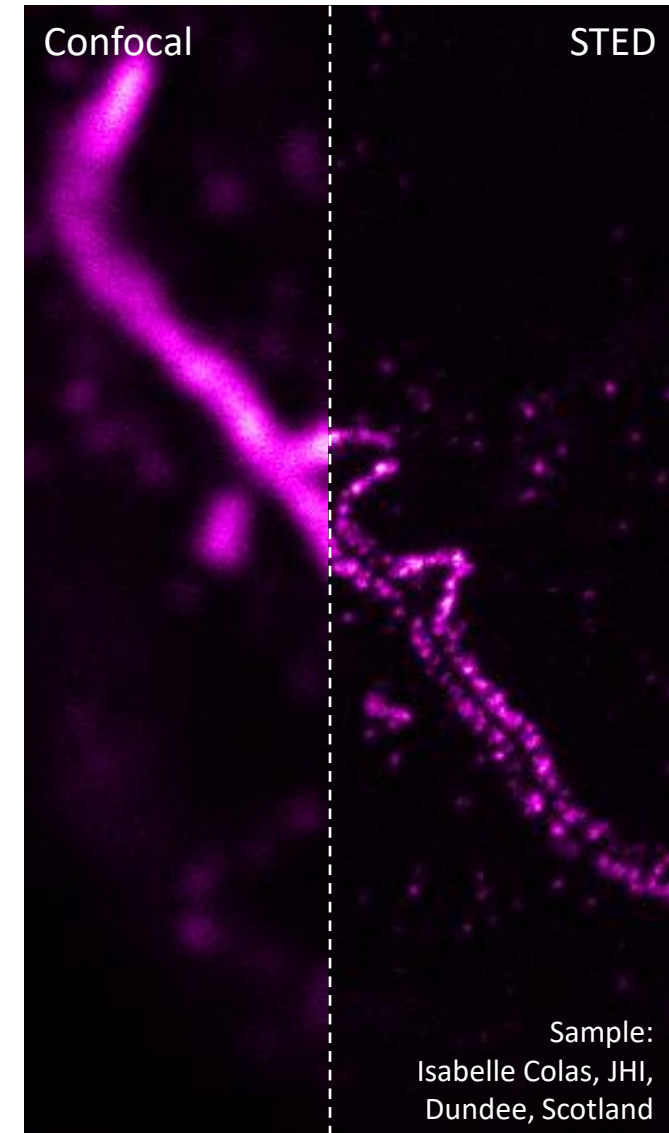


STED: What about... Fixed plant samples



Fixed cell

- Good results quickly achievable
- Use organic dyes for best STED super-resolution
 - Immunostaining (antibodies and/or nanobodies)
 - abberior actin stains (e.g. STAR RED Phalloidin)
 - abberior DNA stains: LIVE 560 DNA, LIVE 590 DNA (works in live-cell and fixed samples)
 - Fixable membrane stain: MemBrite Fix 640/660
- Mounting medium with high refractive index & antifade for best results (see following pages)
- If chloroplast background is expected (e.g. leaves):
 - Chloroform extraction
 - Orange dyes show lower background than red dyes
- See previous pages for more sample and dye information.

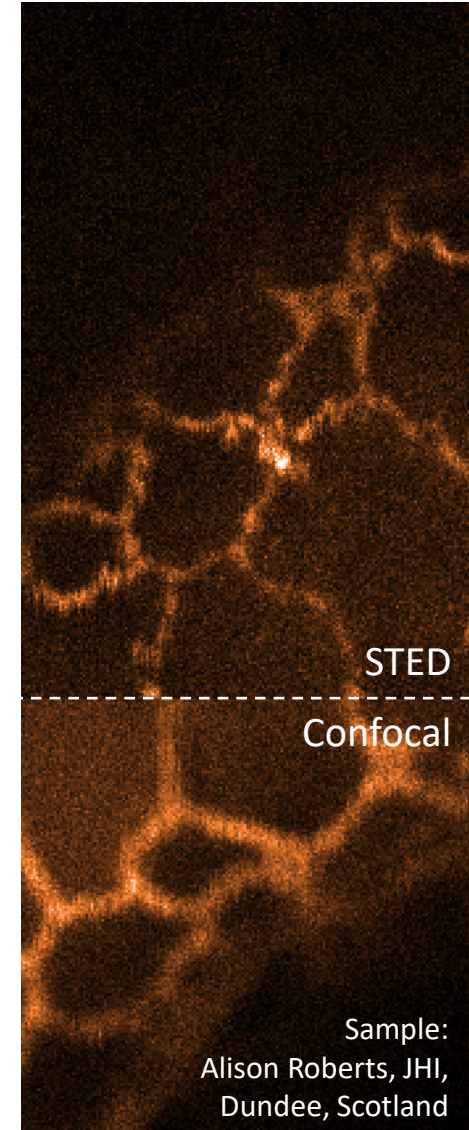


STED: What about... Live plant samples



LIVE cell

- Results can vary depending on species and tissue.
- Autofluorescence: May be reduced via spectral detection & time-gating
- Photodamage
 - Reduce damage: STED 775 nm with orange/red dyes. Image root samples.
- Permeability: Only a few of the tested organic dyes are taken up by live plants:
 - abberior LIVE 560 DNA (and other colours)
 - abberior STAR RED / ORANGE membrane
 - MemBrite Fix 640/660 (also shows plasmodesmata)
- Some classic stains, have also worked well with STED microscopy:
 - FM4-64 (membrane, vacuole), propidium iodide (PM, internal membranes), Nile Red (membranes)
- Dyes for SNAP and Halo tags are available, but permeability needs to be tested.
- Fluorescent proteins: Lower performance than organic dyes.
- If you have stains that perform well in plants, please let us know.



More information online...



Happy labeling & imaging !

For questions contact application@abberior-instruments.com



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